CAUTION **RESEARCH EXPERIMENTATION & LIABILITY DISCLAIMER**

This software is purely research experimentation and is provided strictly "AS IS". It may damage your device, CNC machine, spindle, laser diode, workpiece, or computer. The goal of this project is the exploration of what is possible in unified in-browser digital fabrication. Users accept all risks and agree to hold authors, contributors, and developers completely harmless regardless of outcome. **USE AT YOUR OWN RISK.**

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Community Rules:

- **Open Source & Copyleft:** Keep all innovations accessible and free for everyone.
- **Constructive Community:** Keep all commentary positive, helpful, and respectful.
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JSVCLE Studio — User Guide

JumpStart 3D CAD + Vellum Vector Studio + Continuous Layer Extension (CLE)

Version 01.00.66 — Windows 11 Edition

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1. Overview: What is JSVCLE Studio?

JSVCLE Studio is a unified, all-in-one digital fabrication workspace designed for makers, CNC machinists, woodworkers, and laser engravers. It combines three powerful engineering applications into a single synchronized desktop environment:

1. **JumpStart 3D CAD (`Dev/index.htm`)**: An intuitive, precision boundary-representation (B-Rep) 3D parametric CAD modeling application for mechanical design, solids, extrusions, revolving, and Boolean operations.
2. **Vellum Vector Studio (`Vellum/index.htm`)**: A dedicated 2D vector graphic and laser studio featuring curve drawing, text formatting, path booleans, DXF/SVG import/export, and automatic raster-to-vector tracing.

3. **Continuous Layer Extension (CLE)**: A real-time CAM, multi-axis machine control, and slicing suite directly integrated into JumpStart 3D. CLE features:
    - **Hardware Telemetry & Control:** WebSerial USB and WebSocket networking for FluidNC CNC/Laser controllers with a live 4-axis Digital Readout (DRO).
    - **Interactive Spoilboard & Bed Manager:** Visual CNC table with automatic workpiece seating, breakthrough safe margins, and click-and-drag spoilboard placement.
    - **4th Axis Rotary (A) Chuck & Center Tool:** Live visual chuck, stock cylinder, auto-clamping, and lathe turning tools.
    - **Grayscale to Continuous Z Slicer:** 2D photo-to-3D relief depth extraction with dynamic multi-axis raster toolpath generation.
    - **Machine State Git Vault:** Automatic state snapshotting and version control for machine calibration, tool libraries, and project history.

![ug_architecture.svg](diagrams/ug_architecture.svg)

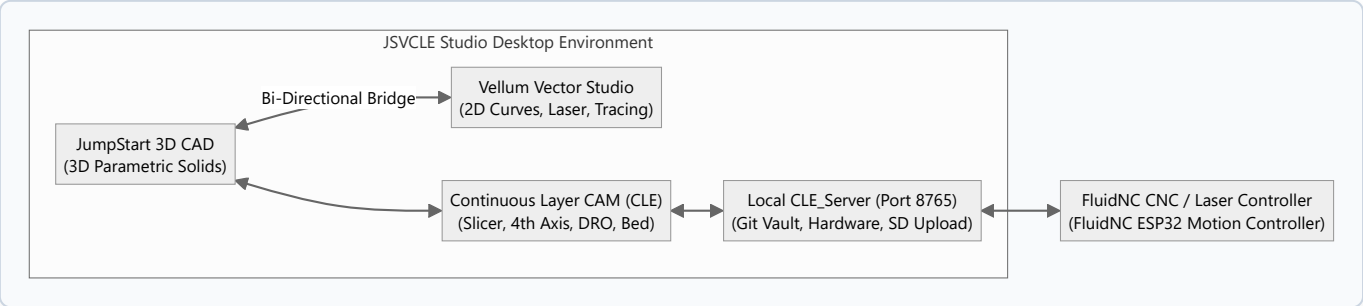


FIGURE: UG ARCHITECTURE

2. System Requirements

Before running the installer, ensure your workstation meets the following specifications:

Component	Minimum Requirements	Recommended Requirements
Operating System	Windows 11 (64-bit, Version 22H2+)	Windows 11 Pro (64-bit, latest release)
Processor (CPU)	Intel Core i7 (8-Way / 8 Cores) or AMD Ryzen 7	Intel Core i9 (24-Way / 24 Cores @ 4.0 GHz+) or AMD Ryzen 9
System Memory (RAM)	16 GB DDR4/DDR5	32 GB+ High-Speed DDR5
Storage (SSD)	2 GB free on High-Speed SATA/NVMe SSD	Ultra-Fast PCIe Gen 4/5 NVMe SSD
Graphics (GPU)	WebGL 2.0 compatible (Intel Iris Xe or dedicated)	NVIDIA GeForce RTX 3060+ / AMD Radeon RX 6700+
Display Resolution	1920 × 1080 (Full HD)	2560 × 1440 (2K QHD) or 3840 × 2160 (4K UHD)
Ports	1× USB 2.0/3.0 port (for CNC USB Serial communication)	USB 3.0 + Wi-Fi 6 (for FluidNC WebSocket)

3. Installation Guide

![ug_installer_flow.svg](diagrams/ug_installer_flow.svg)

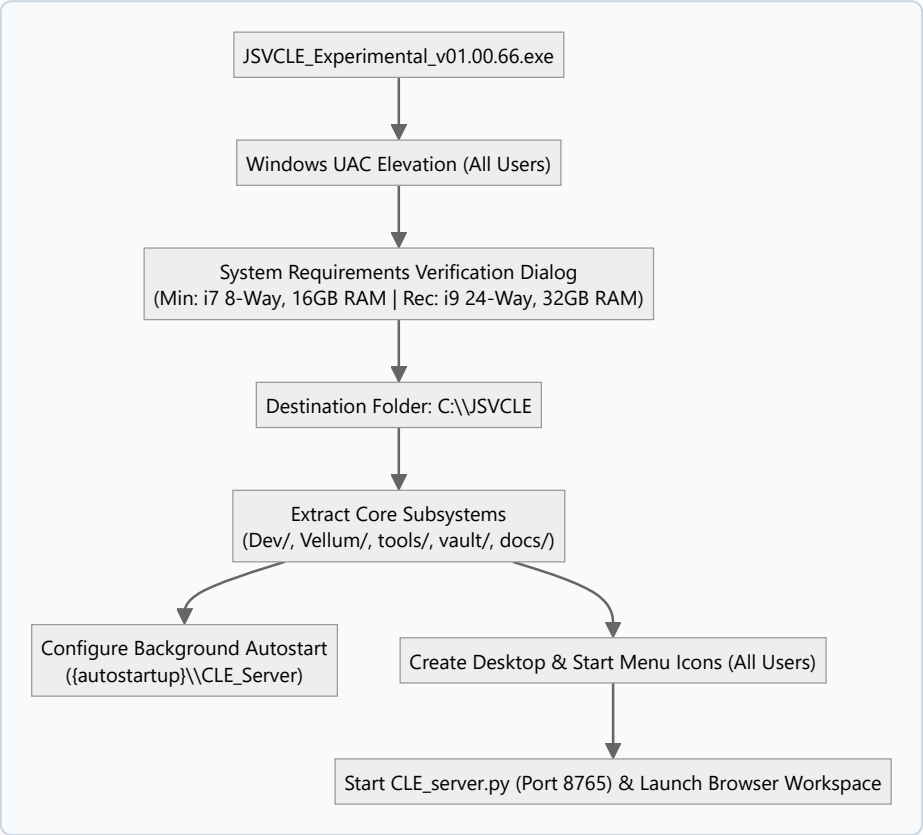


FIGURE: UG INSTALLER FLOW

Step 1: Running the Setup Executable

1. Locate [`JSVCLE_Experimental_v01.00.66.exe`](file:///d:/CLEJS/dist/JSVCLE_Experimental_v01.00.66.exe) in your distribution folder or download location.
2. Double-click the `.exe` installer. If prompted by Windows User Account Control (UAC), click **Yes** to allow installation for All Users.

Step 2: System Requirements Notice

The installer presents a **System Requirements Notice** dialog detailing the minimum (i7 8-way, 16GB RAM) and recommended (i9 24-way, 32GB RAM) specifications. Click **OK** to begin configuration.

Step 3: Destination Location

By default, JSVCLE Studio installs to:

```
C:\JSVCLE
```

You can click **Browse...** to choose any other drive or directory on your system.

Step 4: Installation & Background Service Setup

Click **Install**. The installer automatically extracts:

- `JumpStart 3D and the CLE suite (Dev/)`
- `Vellum Vector Studio (Vellum/)`
- `Local Background Server and Embedded Python Runtime (tools/)`
- `Machine Vault and Tool Library (vault/)`
- `Comprehensive Documentation (docs/)`
- `Windows startup entries in {autostartup}\CLE_Server so the hardware service starts quietly whenever you log into Windows.`
- Desktop and Start Menu shortcuts for all users.

Step 5: Launching the Application

On the final wizard page, leave **Launch JSVCLE Studio** checked and click **Finish**. The background server starts automatically, and the application opens in your default browser.

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4. Initial Configuration

![ug_initial_config.svg](diagrams/ug_initial_config.svg)



FIGURE: UG INITIAL CONFIG

4.1 Connecting to Your Machine

1. Open **JSVCLE Studio**.
2. Click the **CLE Suite** tab in the top workspace bar.
3. In the ribbon under **Connection**:

- **USB Serial:** Click **USB List**, select your machine's COM port (e.g. COM3 for CH340 / CP2102), and choose baud rate 115200.
- **WiFi / WebSocket:** If running FluidNC over Wi-Fi, click **WiFi/WS** and enter your controller's IP address (e.g. 192.168.1.150).
- Once connected, the top status badge will turn green: FluidNC: Idle.

4.2 Configuring the Machine Bed & Spoilboard

1. Click **Setup & Spoilboard** in the CLE ribbon.
2. Select your machine profile from the dropdown (e.g. **FluidNC Standard CNC / Laser** or custom machine profile).
3. The 3D viewport instantly renders your exact spoilboard dimensions (e.g. $600 \times 500 \times 20\,\text{mm}$ MDF), aluminum extrusion base, and live origin reticle.
4. If your spoilboard thickness differs from standard ($20.0\,\text{mm}$), enter your measured thickness in the Spoilboard dialog and click **Save to Vault**.

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5. Three Real-World Usage Workflows

![ug_three_workflows.svg](diagrams/ug_three_workflows.svg)

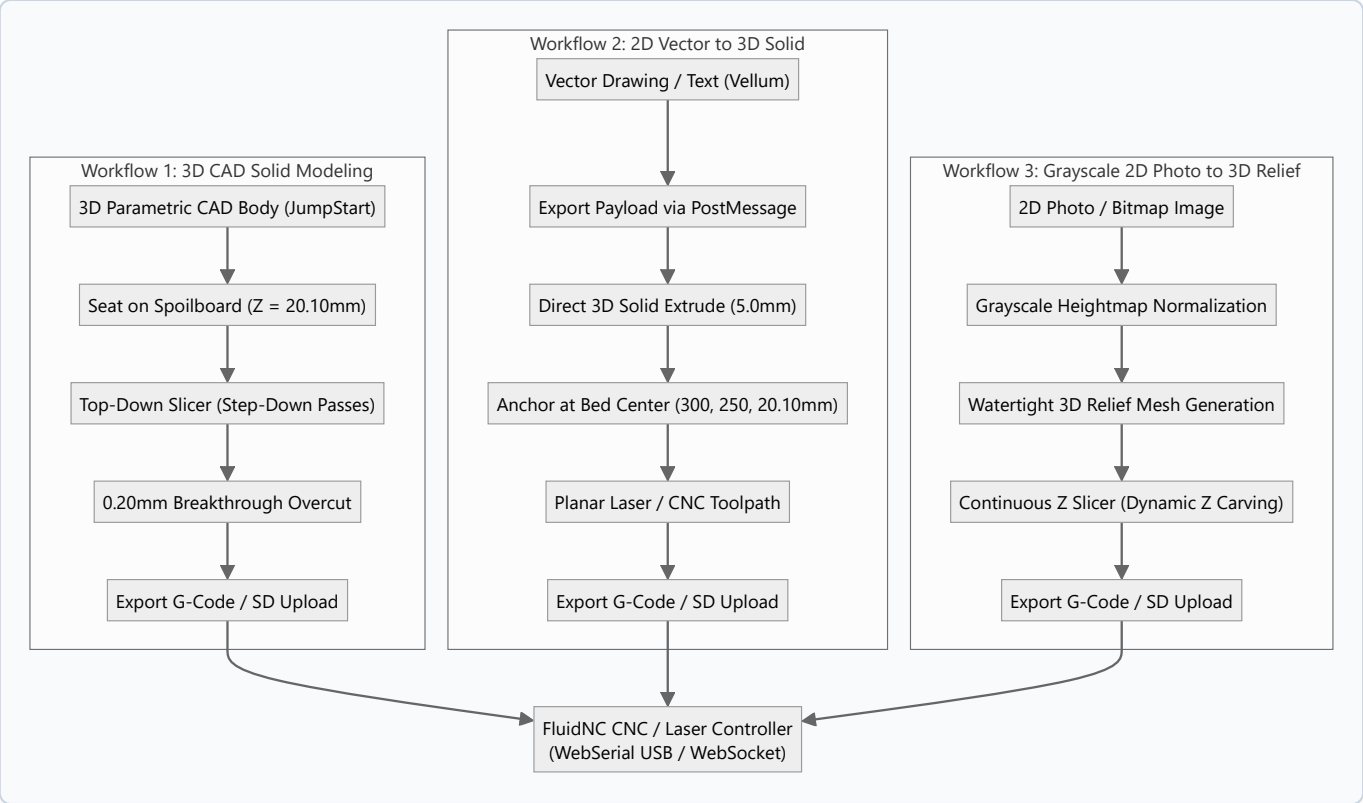


FIGURE: UG THREE WORKFLOWS

Example 1: 3D CAD Solid Modeling & Continuous Layer Slicing

Goal: Model a mechanical mounting bracket and generate a multi-pass CNC carving toolpath.

1. **Design the 3D Part:**
   - In the **Design 3D** workspace tab, click **Box** or draw a 2D sketch and click **Extrude** (e.g., $100 \times 60 \times 15\,\text{mm}$).
   - Add mounting holes using the **Cylinder (Hole)** tool.
   - **Place on CNC Spoilboard:**
   - Switch to the **CLE Suite** tab.
   - Click the 3D body on the bed. A luminous neon cyan outline and the **Bed Placement HUD** appear.
   - Click 🎯 **Center** on the HUD to position the bracket symmetrically at $(X=300, Y=250)$ on the spoilboard.
   - **Slice to Toolpaths:**
   - In the ribbon, click **Slice & 3D Stack**.
   - Select your Target Machine: **CNC Milling (Top-Down passes)**.
   - Set Layer Height (e.g. $1.5\,\text{mm}$ step-down) and Spoilboard Breakthrough ($0.20\,\text{mm}$ overcut for clean part release).
   - Click **Slice & Display in 3D Viewport**.
   - **Simulate & Export:**
   - The interactive **Slicer HUD** mounts at the bottom of the viewport.
   - Click **Play Trace** to watch the simulated toolhead carve each layer in real time.
   - Click **Export G-Code (.nc)** or **Upload to SD Card** to transfer the program directly to your machine.

Example 2: 2D Vector Design in Vellum & Extrusion to 3D

Goal: Create an artistic vector plaque in Vellum Vector Studio and transfer it into 3D CAD.

1. **Open Vellum Vector Studio:**
    - Click the **Vector Studio** button in the CLE ribbon (or click **Vector** in the workspace tabs).
    - Vellum opens in full-screen overlay mode.
    - **Draw Vector Geometry:**
    - Use the **Pen (Curve)**, **Shape**, or **Text** tools to draw your emblem.
    - Use **Autotrace** if importing a bitmap logo (.svg, .png).
    - **Export to JS Workspace:**
    - In Vellum's top menu, click **Export** $\rightarrow$ **Save to JS Workspace...**.
    - Choose **Option 2: 3D Extruded Body (Solid)** and enter extrusion thickness (e.g. $6.0\,\text{mm}$).
    - Click **Transfer to JS Workspace**.
    - **Result:**
    - Vellum closes and returns you directly to JumpStart 3D.
    - Your vector shape is converted into a native 3D watertight solid, resting at bed center $(300, 250, 20.10\,\text{mm})$ on the spoilboard!

Example 3: Grayscale Photo to 3D Relief & Continuous Z Carving

Goal: Convert a 2D portrait photo into a carved 3D wood relief carving.

1. **Import Image:**
    - Click **Grayscale 3D** (#cle-r-grayscale) in the CLE ribbon.
    - Under *Select Image Source*, choose 📁 **Browse File...** (or select a photo already transferred from Vellum).
    - **Modulate Image Depth:**
    - View the live thumbnail on the preview canvas.
    - Adjust **Contrast**, **Brightness**, and **Gamma Curve** to bring out facial contours.
    - Check **Invert Grayscale** if white represents high peaks and dark represents low carving floors.
    - **Set Physical Dimensions:**
    - Width: $120.0\,\text{mm}$, Height: $90.0\,\text{mm}$.
    - Max Relief Depth: $5.0\,\text{mm}$, Base Floor: $2.0\,\text{mm}$.

- **Choose Action:**
- Click 📦 **Create 3D Object in Workspace**: Generates a watertight solid 3D mesh (maple finish) in the CAD scene for parametric combination.
- Click ⚡ **Slice to Continuous Z**: Generates smooth multi-axis 3D carving raster toolpaths that vary Z dynamically along each pass, and mounts the Slicer HUD for simulation!

6. Troubleshooting Guide

Symptom	Probable Cause	Recommended Solution
FluidNC Offline / COM Port not found	USB cable disconnected or CH340 driver missing	Check USB connection. In Windows Device Manager, ensure the COM port shows under *Ports (COM & LPT)*. Install the CH340/CP2102 driver if needed.
"CLE_server not responding" warning	Background Python service stopped	Run `cle_server_start.bat` from your install directory (`C:\JSVCLE`). Verify that `http://127.0.0.1:8765/api/status` returns `{"status":"running"}` in your browser.
3D Viewport is blank or black	WebGL disabled or hardware acceleration off	In your browser, go to `chrome://settings/system` (or edge settings) and enable **Use graphics acceleration when available.** Verify WebGL support at `get.webgl.org`.
Slices cut through the aluminum bed	Zero point misconfigured or no spoilboard defined	Ensure your spoilboard thickness is entered correctly in the Spoilboard Setup modal ($20.0\,\text{mm}$). Never set cutout overcut greater than spoilboard thickness.
Vellum "Save to JS Workspace" doesn't switch	Browser popup or cross-frame permission blocked	JSVCLE uses standard internal frame messaging. Ensure you are launching the application via `cle_start.bat` or `localhost` rather than a restricted sandbox.

7. Uninstallation

To remove JSVCLE Studio completely from your computer:

1. Open **Windows Settings** $\rightarrow$ **Apps** $\rightarrow$ **Installed Apps.**
2. Locate **JSVCLE Studio** in the list.
3. `Click the three dots ...` and select **Uninstall.**

(Alternatively, run `C:\JSVCLE\unins000.exe` directly or click "Uninstall JSVCLE" in the Start Menu).

1. The uninstaller removes all desktop icons, start menu folders, startup background service hooks, and application files cleanly.

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8. Glossary of Terms

- **DRO (Digital Readout):** Real-time display of machine axes (X, Y, Z, A) in Work Coordinate System (WCS) millimeters and degrees.
- **WCS (Work Coordinate System):** The part coordinate frame (`G54`), where $(0, 0, 0)$ is set relative to your workpiece stock rather than machine limits.
- **Spoilboard:** A sacrificial MDF board mounted on top of the aluminum extrusion bed that allows through-cuts without damaging cutting tools or the machine table.
- **Continuous Z Slicing:** A 3D CAM carving strategy where the toolhead moves smoothly in $X, Y,$ and Z simultaneously following natural surface height contours, avoiding stepped stair-stepping.
- **4th Axis (A-Axis):** A rotary chuck and tailstock alignment tool that replaces the spoilboard for round stock turning, fluting, and multi-sided indexing.
- **Breakthrough Pass:** A final perimeter cutout pass that cuts $0.20\,\text{mm}$ into the MDF spoilboard so parts release cleanly without tabs.
- **FluidNC:** High-performance, open-source CNC and laser motion control firmware running on ESP32 microcontrollers with WebSerial and WebSocket support.